Detecting Fraudulent Job Postings: Exploratory Analysis and Detection Strategy

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1 Problem Context

Online job platforms face increasing risks from fraudulent job postings that exploit job seekers. This analysis explores the EMSCAD dataset containing 17,880 job advertisements labeled as legitimate or fraudulent. This summary report identifies practical signals that can help job platforms detect fraudulent postings early and flag suspicious listings for review.

2 Dataset Audit

The dataset contains 17,880 job postings, of which 866 (4.8%) are labeled as fraudulent. This indicates a highly imbalanced classification problem. The dataset contains a mixture of text fields (e.g., job descriptions and requirements), structured job attributes (e.g., employment type and industry), and platform metadata (e.g., presence of company logos and screening questions).

3 Key Insights from Exploratory Analysis

3.1 Short company profiles are more likely to be fraudulent

- Fraudulent postings are heavily concentrated around near-zero company profile text lengths. In contrast, legitimate postings show a much wider range of profile lengths and often include longer company descriptions. (Figure 3.1a)
- Description lengths appear similar between the two groups. (Figure 3.1b). However, fraudulent postings tend to have requirements shorter than about 500 characters. (Figure 3.1c)
- Hence, scammers invest less effort in describing the employer and the qualifications needed for the role, making company profile presence and requirements length useful signals for detecting fraud.

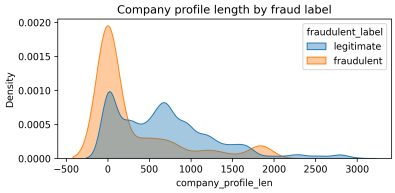

Figure 3.1a: Comparing company profile text length to fraud

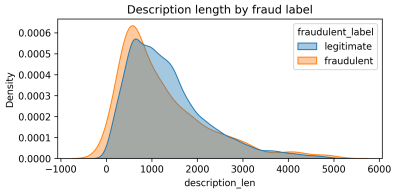

Figure 3.1b: Comparing description text length to fraud

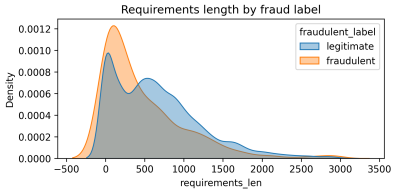

Figure 3.1c: Comparing requirements text length to fraud

3.2 Missing platform credibility signals such as logos, remote work, screening questions correlate with fraud.

- Job listings without company logos have a fraud rate of about 16%, compared with roughly 2% for postings that include a logo. (Figure 3.2a)
- Similarly, listings without questions have a fraud rate of around 7%, more than double the roughly 3% fraud rate for postings that require applicant responses. (Figure 3.2b)
- Remote postings also show higher fraud rates (about 8%) compared with non-remote postings (around 5%). (Figure 3.2c)
- Overall, postings that lack company information or platform verification features are much more likely to be fraudulent.

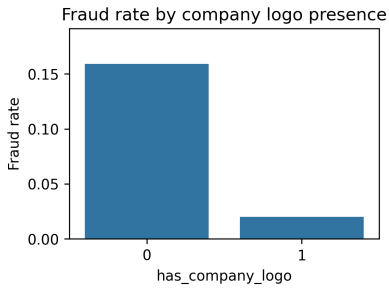

Figure 3.2a

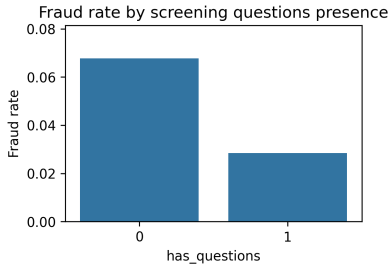

Figure 3.2b

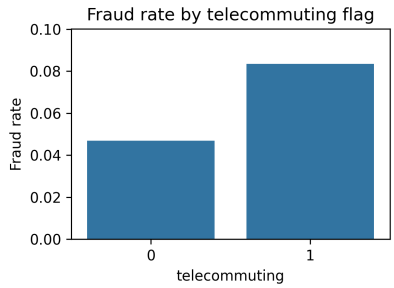

Figure 3.2c

3.3 Recurring patterns and signals

Fraudulent postings often contain shorter requirements sections and vague job descriptions, indicating potential linguistic signals useful for detection.

3.4 Challenges and Limitations

Several characteristics of the dataset present challenges for the proposed approach.

- **Class imbalance.** Fraudulent postings represent only about 4.8% of the dataset. Evaluation should therefore focus on metrics such as precision, recall, and F1 score rather than accuracy.
- **High-dimensional text features.** Many predictive signals come from textual fields such as job descriptions and company profiles. A simple and interpretable classification model can serve as a strong baseline, with more advanced models explored later if needed.
- **Evolving fraud patterns.** The dataset contains job postings collected between 2012 and 2014. Scam tactics may evolve over time, so a real-world system would need to be retrained regularly using more recent data.
- Despite these limitations, the dataset still provides strong signals for identifying fraudulent postings, particularly through employer credibility indicators and platform-level metadata.

4 Proposed Detection Approach

A hybrid fraud detection approach combining textual features and structured metadata is recommended. Key feature groups include:

- Employer credibility signals (company profile presence and length)
- Text features (TF-IDF representations) of job descriptions and requirements, text length indicators
- Platform metadata such as `has_company_logo`, `has_questions`, and `telecommuting`

A logistic regression model with text features provides a strong and interpretable baseline, with tree-based models (e.g., Random Forest or Gradient Boosting) as potential extensions for non-inear relationships

5 Evaluation and Deployment Strategy

Because fraud detection is a risk management problem, evaluation should focus on detecting scams while minimizing false alarms.
Model performance would first be evaluated using a hold-out validation set (80/20, 70/30) or stratified cross-validation to ensure reliable results despite the dataset class imbalance.
Given the class imbalance, evaluation should focus on:

- Precision - proportion of flagged postings that are actually fraudulent
- Recall - ability to detect fraudulent postings
- F1 score - balance between precision and recall

Additionally, fields such as `in_balanced_dataset` should not be used as model features. This variable is related to how the dataset was constructed rather than information available at prediction time, and including it could introduce data leakage.
If the model consistently detects a high proportion of fraudulent postings while keeping false positives at a manageable level, it would be suitable for pilot deployment.

6 Operational deployment

A realistic system would function as a risk scoring tool:

Job posting submitted → Feature extraction → Fraud risk score → High-risk postings flagged for manual review

This approach allows human moderators to review suspicious postings while minimizing disruption to legitimate employers.